

Mixed Reality Visualization of Room Impulse Response Map using Room Geometry and Physical Model of Sound Propagation

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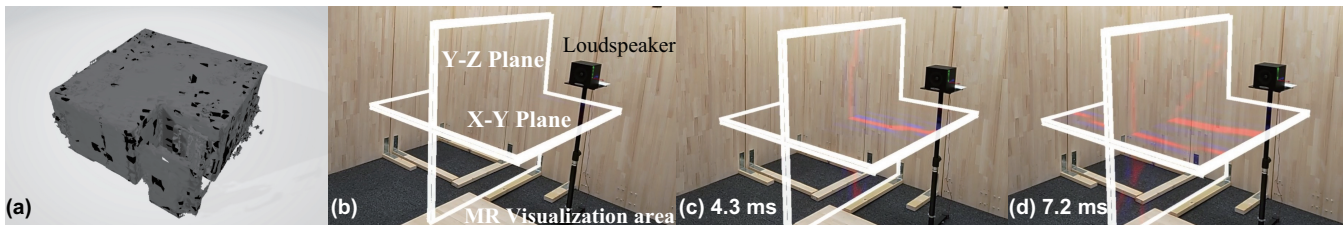


Figure 1: MR visualization of the room impulse responses. (a) Room shape obtained using SLAM. (b) Visualization setting using MR. (c) and (d) Snapshots of sound propagation after animating the RIR map at 4.3 and 7.2 ms. The red and blue colors represent the positive and negative sound pressures, respectively.

ABSTRACT

In this paper, an MR visualization method based on sound field modeling is proposed. Using a small quantity of measurement data, the sound field was modeled using equivalent sources and room shapes acquired via SLAM. From the modeled sound field, the estimated room impulse responses at the target grid points were then animated to visualize the sound field using MR technology. Consequently, the animation of the sound field in MR clearly represented how sound propagates, including reflections.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality.**

KEYWORDS

Mixed Reality, Spatial mapping, Interpolation, Sound propagation

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1 INTRODUCTION

One of the applications of Mixed Reality (MR) technology is the visualization of scientific data such as electromagnetic waves and sound fields. In particular, because sound propagation is closely related to boundary conditions such as the shape of rooms and objects, MR visualization is effective for understanding the relationship between sound propagation and the boundary conditions to solve noise problems and design the room acoustics. For steady sound fields, Inoue *et al.* proposed a real-time visualization system for 3D sound intensities using MR and a handy microphone array [Inoue *et al.* 2019].

For transient sound fields driven by a loudspeaker, Watanabe *et al.* proposed an MR visualization system based on the spatial interpolation of room impulse responses (RIR) from the moving path of a handy microphone [Watanabe *et al.* 2020]. However, there were problems with the interpolation accuracy because it was significantly degraded by errors in the microphone's moving path.

In this study, an MR visualization method for sound propagation based on sound field modeling is proposed to improve the visualization accuracy of the transient sound field and reduce the measurement time. Using RIRs with a smaller number of fixed microphones, the sound field was visualized by interpolating RIRs based on the sparse equivalent source method and the room shape acquired via simultaneous localization and mapping (SLAM).

2 PROPOSED METHOD

Figure 2 illustrates the concept behind the proposed method. A map of the RIRs was visualized based on sound field modeling from a smaller number of RIRs [Tsunokuni *et al.* 2021]. It was assumed that the positions of the sound source and microphones

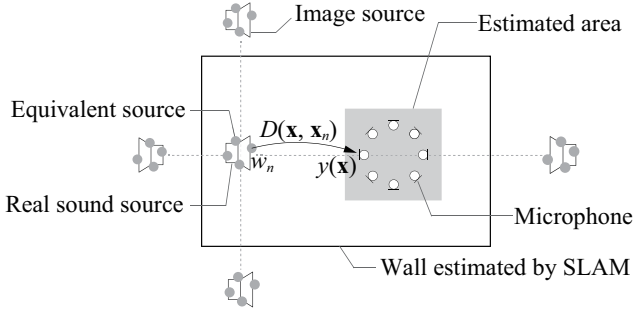


Figure 2: Concept of the proposed method. The RIRs in the target sound field are modeled via the superposition of sparse equivalent sources. The equivalent sources are then placed around the approximate positions of the sound and image sources. Subsequently, the positions of the image sources are calculated from the room shape using SLAM based on geometrical acoustics.

are approximately known, and the room shape was estimated from the outline of a spatial map using SLAM. Based on the equivalent source method [Koopmann et al. 1989], the target sound field was represented via the superposition of equivalent sources around both the sound and image sources based on geometrical acoustics. In the frequency domain, the transfer function y in the target sound field $\mathbf{x}$ is expressed as follows:

$$y(\mathbf{x}) = \sum_{n=1}^N D(\mathbf{x}, \mathbf{x}_n) w_n, \quad (1)$$

where $\mathbf{x}_n$ and w_n are the position and weight of the n -th equivalent source, respectively. Additionally, D is the transfer function of the equivalent source, which is Green's function of the point source in a three-dimensional free field.

Here, Eq. (1) holds for all microphones because the microphone positions $\mathbf{x}'_m$ are located in the target area. The transfer function vector $\mathbf{y}$ for M microphones is represented by

$$\mathbf{y} = \mathbf{D}\mathbf{w}, \quad (2)$$

where $\mathbf{y} = [y(\mathbf{x}'_m)]_{m=1,\dots,M}$ is the transfer function vector of microphones. $\mathbf{D} = [D(\mathbf{x}'_m, \mathbf{x}_n)]_{m=1,\dots,M, n=1,\dots,N}$ is the transfer function matrix between equivalent sources and microphones. $\mathbf{w} = [w_n]_{n=1,\dots,N}$ is weight vector of equivalent sources. Because the vector of the weight coefficients $\mathbf{w}$ is unknown, the inverse problem must be solved. Thus, it was assumed that sparse equivalent sources are required to represent the sound field because there is only one sound source and the number of image sources is sufficiently small. The following optimization problem was solved to obtain the weight vector $\mathbf{w}$:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{minimize}} \quad \|\mathbf{w}\|_1 \\ & \text{subject to} \quad \|\mathbf{y} - \mathbf{D}\mathbf{w}\|_2 \leq \epsilon \end{aligned} \quad (3)$$

where $\|\cdot\|_1$ and $\|\cdot\|_2$ are ℓ_1 -norm and ℓ_2 -norm, respectively, and ϵ is the error tolerance. Subsequently, the transfer function at any point was estimated using Eq. (1). To obtain the animation of the sound field, the transfer functions in the frequency domain were converted to RIRs in the time domain using an inverse FFT.

3 EXPERIMENT

A simulation experiment in which the sound field was visualized in a three-dimensional using two planes was conducted. As shown in Figure 1(b), the sound field in front of the loudspeaker was visualized in a soundproof room. 60-ch cylinder microphone array was positioned in the target sound field, which includes both direct and first-order reflections. Consequently, a target frequency range from 0.5–4 kHz and sampling frequency of 48 kHz were obtained. A wavefront was also visualized in real space using Microsoft HoloLens2, as shown in Figure 1(c) and (d).

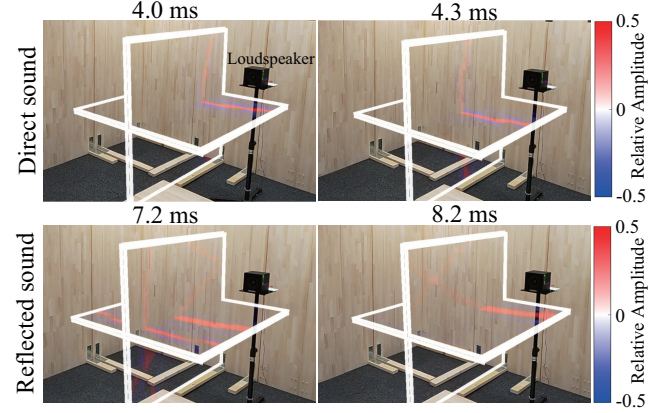


Figure 3: MR visualization of the 3D RIRs. The size of the target sound field is 1.0 m × 1.0 m × 1.0 m; and we visualized two planes, the X-Y plane and the Y-Z plane at the center of the speaker. The red and blue colors represent the positive and negative amplitude of the RIRs, respectively.

Figure 3 presents the simulation results. The sound pressure distribution was converted into 2D colored graphics to visualize the flow of the wavefronts as an animation. The figures show direct sound propagation from the front of the loudspeaker at 4.0 and 4.3 ms. At 7.2 and 8.2 ms, the visualized sound fields indicate that the reflected sounds are propagated from floor and ceiling.

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